496 Software Project:
CSD Central

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2026-04-17

1 Client Information

By sharing this client information and the rest of this document, you are stating that this client has provided this project as something they want (not something you created and asked if they wanted), and that they are interested in having you complete this project for your capstone.

- Client name: Sandy Banthem
- Client title: Director of Operations and Finance
- Client email address: sandybanthem@chesarda.org
- Client employer: Chesapeake Search Dogs
- How you know the client: A student at Loyola introduced us.

2 Project Description

2.1 Overview

We were tasked to create an internal website for Chesapeake Search Dogs for communication and organization for internal staff. This project is streamlining internal organization for employees to increase efficiency.

2.2 Key Features

Some key features our client needs are a homepage, calendar, certification handling, log handling, and users.

The homepage must have a list of upcoming trainings, RSVPs, and weather forecast for that day. The website theme must also have navy blue as its main color, this was specifically requested by the client. It will not include the calendar.

This will be a seperate page. A calendar is also necessary to efficiently view upcoming RSVPs and scheduling work. User roles affects how a user can interact with the calendar. Admins can add trainings with info, date, time, and location. Each registered user can RSVP for an event on the calendar with a yes, no, or maybe. They also have the option to add a note. Every user in the system can see other's RSVP to coordinate accordingly. Each event also has communication logs with user numbers with a time stamp.

Certifications are important for the client and they need a way for users to send in their certifications. It will be sent as an image with the user name, date, date of certification, option of expiration, and expiration date to a database. Admins can view all certifications and users can only modify and view and their own certifications.

The client also needs a system to track and see logs for the users. Users can keep track of their training logs and they are connected to the calendar trainings. The date, location, time, weather, start and stop times are automatically filled in from the training. Users can also select a dog from their dog list.

Now, finally, the client needs a way to manage users. basic user control and security will be implemented. Users will have the parameters: name, email, username, password, birthday, address, phone, CSD Number, emergency contact, and emergency phone. They will also be able to enter dogs if applicable. Dogs will have different parameters: name, DOB, vet, status, color.

2.3 Why this Project is Interesting

We like their cause, they work with first responders and help save lives. Since we are building it from scratch as two people, it seems like a fair challenge. We have to consider real-world factors like security, backups, website uptime/reliability, and real user experience. And to add to the challenge, the client only has a rough idea of what they want right now, naturally we will probably have to communicate and change or add features to their liking throughout the process.

2.4 Areas of CS required

The areas of Computer Science required for this project are:

- Software Engineering
- Databases
- System Design
- Web Programming
- Full-Stack Development
- DevOps Cybersecurity

2.5 Potential Concerns and Questions

Ekto: Some concerns for me are keeping sensitive information secure and a backup database. Salting/hashing or two-factor authentication is probably the best option to keep user accounts secure, but keeping a backup database would be something new for me.

Alberto: I am optimistic that this project will work. I do believe that the weather aspect to the RSVP is a bit unnecessary, but it should not be too difficult to implement. There are multiple weather APIs to choose from.

2.6 Summary of Efforts to Find a Project

Ekto spoke with the client in November after a classmate introduced us, and they provided us the tentative list of features a few weeks afterwards. They were looking for someone to handle the development of an internal website. Ekto reached out to a handful of clients prior to this, but never received response.

3 Requirements

3.1 Non-Functional Requirements

Table 1 presents the NFRs for this software project.

ID	NFR Title	Category	Description
NFR1	Navy Blue Theme	Cosmetic	Make website have a Navy Blue theme.
NFR2	Backup DB	Reliability	Create a backup DB for data integrity.
NFR3	Website Deployment	Reliability	Deploy the website to be used by client.
NFR4	Website Reliability	Reliability	Make sure website is designed and created to be robust and reliable.

Table 1: Non-Functional requirements

3.2 Functional Requirements (User Stories)

Table 2 presents the functional requirements for this software project.

ID	Story Title	Points	Description
S1	Viewing Upcoming Events	2	As an user, I want be able to see upcoming events on the homepage, so that I can keep track of upcoming events.
S2	Viewing Event RSVP's	2	As an user, I want to see other people's RSVP status, so that I can see who else will be attending events.
S3	Weather Display On Homepage	2	As an user, I want to see the daily weather on the homepage, so that I can plan for training conditions
S4	Event RSVP's	3	As an user, I want to be able to RSVP Yes/Maybe/No for upcoming events, so that I can show my intention to attend events
S5	Adding Events to Calendar	8	As an admin, I want to manage and posy events to the calendar with info, date, time and location, so that users can plan accordingly events on scheduled days
S6	Viewing Event RSVP's (Again)	2	As an user, I want to see other people's RSVP status on the calendar, so that I can see who else will be attending events
S7	Live Calendar	5	As an user, I want to have a live calendar, so that I can view scheduled events
S8	Event Notes	2	As an user, I want to be able to leave notes for events, so that I can plan for myself
S9	Upload Certification	3	As an user, I want to be able to upload my certifications as an image and fill out name, date achieved, expiration details, and add a title for the upload, so that I can be certified
S10	View Certifications	1	As an user, I want to be able to view my certifications, so that I can see what certifications I have
S11	Manage Certifications	2	As an user, I want to be able to modify or delete certification details, so that I can keep my certifications up to date
S12	Admin Viewing Certifications	1	As an admin, I want to see my users certifications, so that I can make sure my team is certified
S13	Certifications Filter/Search	5	As an user, I want to be able to filter certifications by date added, recently edited, or search by title name, so that I can quickly lookup my certifications
S14	Communications Logs	3	As an user, I want to be able to fill out communcations logs from events, so that I can keep track of what happened at events
S15	Training Logs	5	As an user, I want to be able to fill out training logs with date, time, weather automatically filled in from selected scheduled calendar event, so that I can easily fillout trainings
S16	Adding Dog To Training Logs	2	As an user, I want to be able to add a dog to training logs, so that I can keep track of their progress
S17	Editing Logs	2	As an user, I want to be able to edit/delete logs, so that I can make sure they are filled out correctly and accurate

S18	Logs Filter/Search	5	I want to be able to filter logs by date added, recently edited, or by type of log
S19	Creating Users	2	As an user, I want to be able to create an account with username, password, name, email, birthdate, address, phone number, CSD number, emergency contact, and emergency phone, so that I can use the platform
S20	Creating A Dog Account	2	As an user, I want to be able to create an account for my dog with their name, birthdate, vet, status, and color, so that I can use the dogs for logs
S21	Admin Control	8	As an admin, I want to be able to have the ability to have full view to registered users/dogs and be able to delete them, so that I can control the registered users
S22	Uploading Documents	3	As an admin, I want to be able to upload documents, so that my team can view important files.
S23	Deleting Documents	1	As an admin, I want to be able to delete documents, so that I can only keep necessary documents.
S24	Viewing Documents	1	As an user, I want to be able to view documents, so that I can view important files.
S25	Documents Filter/Search	5	As an user I want to be able to search documents by name and filter by date ranges, so that I can easily lookup documents.
	Total:	77	

Table 2: Functional requirements as User Stories.

4 System Design

4.1 Architecture

We are using MVC for our project. The figure is also provided to show more concrete examples.

Model: MongoDB schema.

View: React will be used to show UI, user communication, and communicating with API's.

Controller: Bun API's will be used for routing, authentication, and authorization.

4.2 Diagrams

Class diagram for project is shown below.

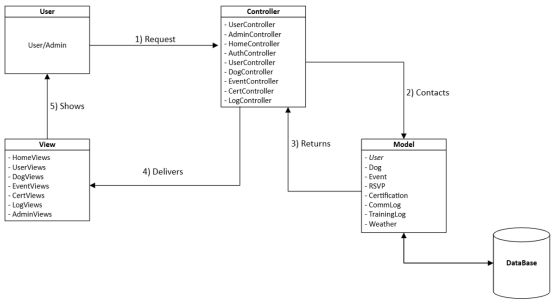


Figure 1: Initial Class Diagram

4.3 Technology

React is used for the front-end. Bun is a modern JS runtime with many integrated features, including testing and API toolkits. MongoDB is chosen because it is NoSQL and requirements from the client may change in the future. Docker is used to containerize the application.

Tech	**Type**	**Usage**
JavaScript	Language	Backend and Frontend
Bun	Runtime	Javascript Runtime Toolkit
Docker	Containerizer	Containerize the app for consistency
React	UI Framework	Used for UI design and implementation
MongoDB	No-SQL DB	Long-term data storage
Bun Test Runner	Testing tool	Used to test JavaScript code
Bun API	API creation tool	Toolkit to create APIs
GitHub Actions	CI/CD Tool	DevOps tool to automate version control for the repository

Table 3: Technologies used in the project

4.4 Coding Standards

We will be using camelCase in our codebase, singular and lowercase names for MongoDB, and RESTful (context, version, resource, parameter) for APIs.

4.5 Data

User: id, name, email, username, password, role, birth date, address, phone, CSD phone, emergency contact name, emergency contact phone, dogs.

Dog: id, owner's id, name, birth date, vet, status, color.

Event: id, title, description, date, start time, end time, location, RSVPs (user, status, note).

Certification: id, user, title, image, date achieved, expiration possibility, expiration date, date uploaded, date modified.

Training Log: id, user, event, date, location, weather, start time, end time, dogs, notes, date modified.

Communication Log: id, event, user, message, time.

4.6 UI Mocks

[Define and draw/sketch/code the main UIs your user will interact with in your software. Add your UI mocks here and a short caption about it. Do not forget about the main forms and CRUD UIs.]

4.7 UI Mocks

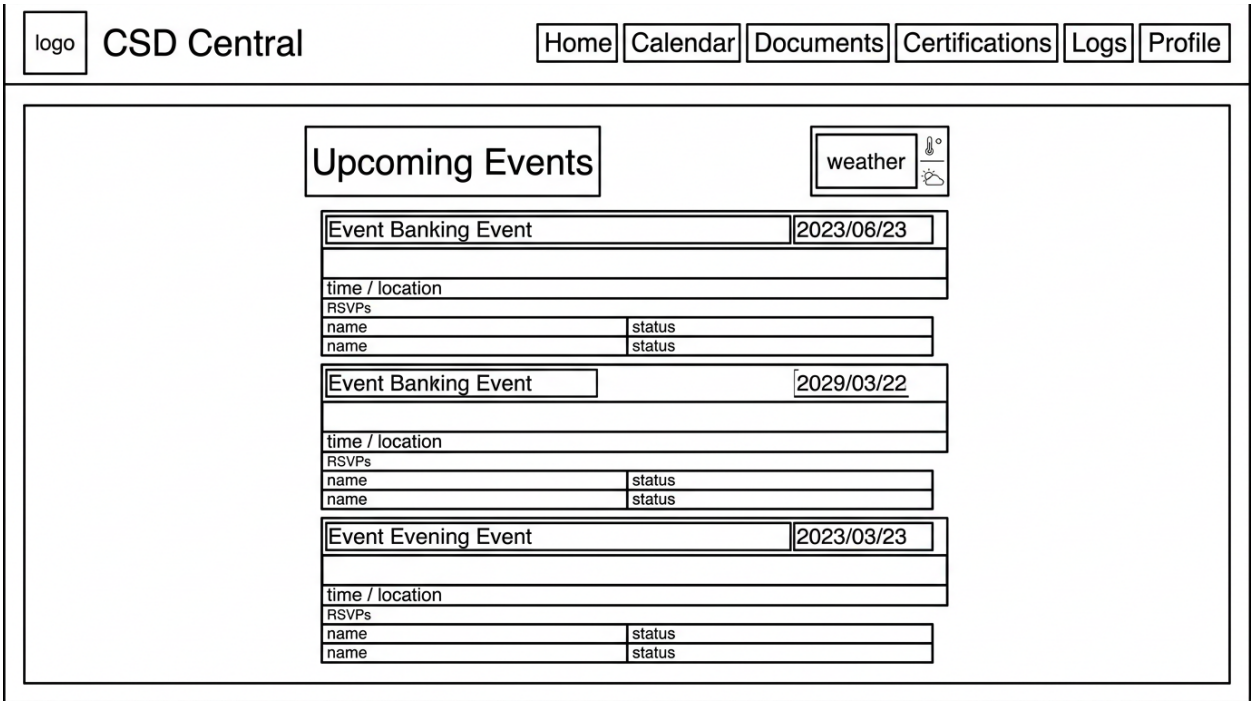


Figure 2: UI Mockup: Homepage

Figure 3: UI Mockup: Login Page

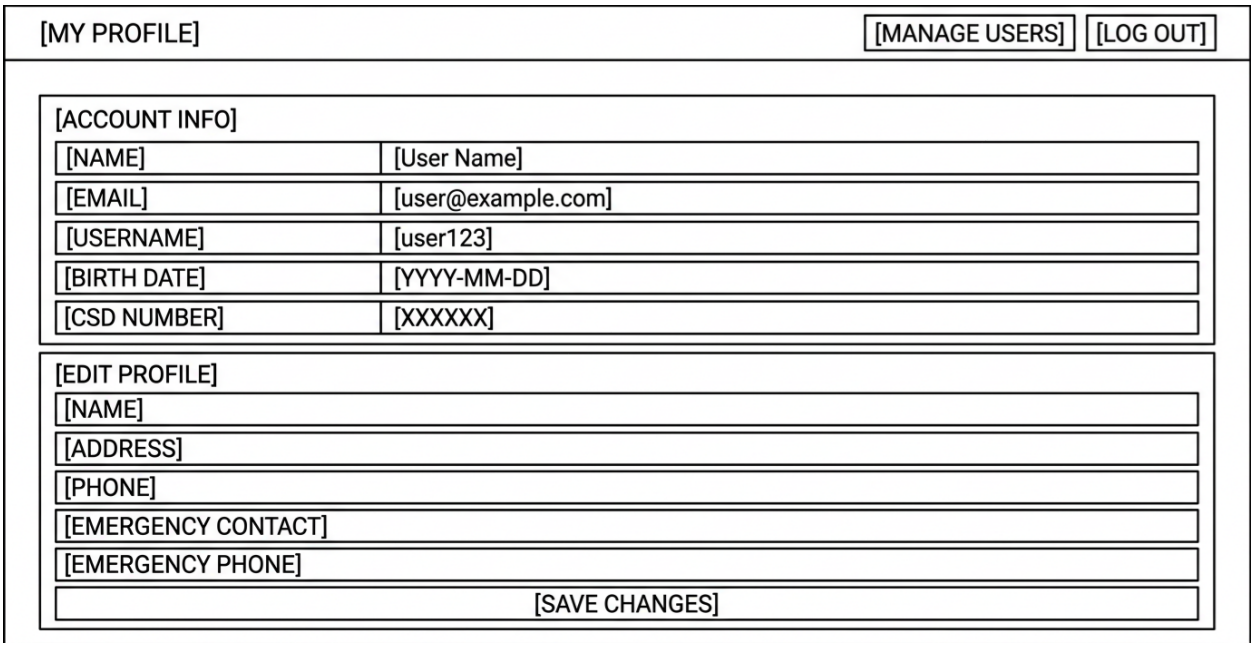


Figure 4: UI Mockup: User Dashboard

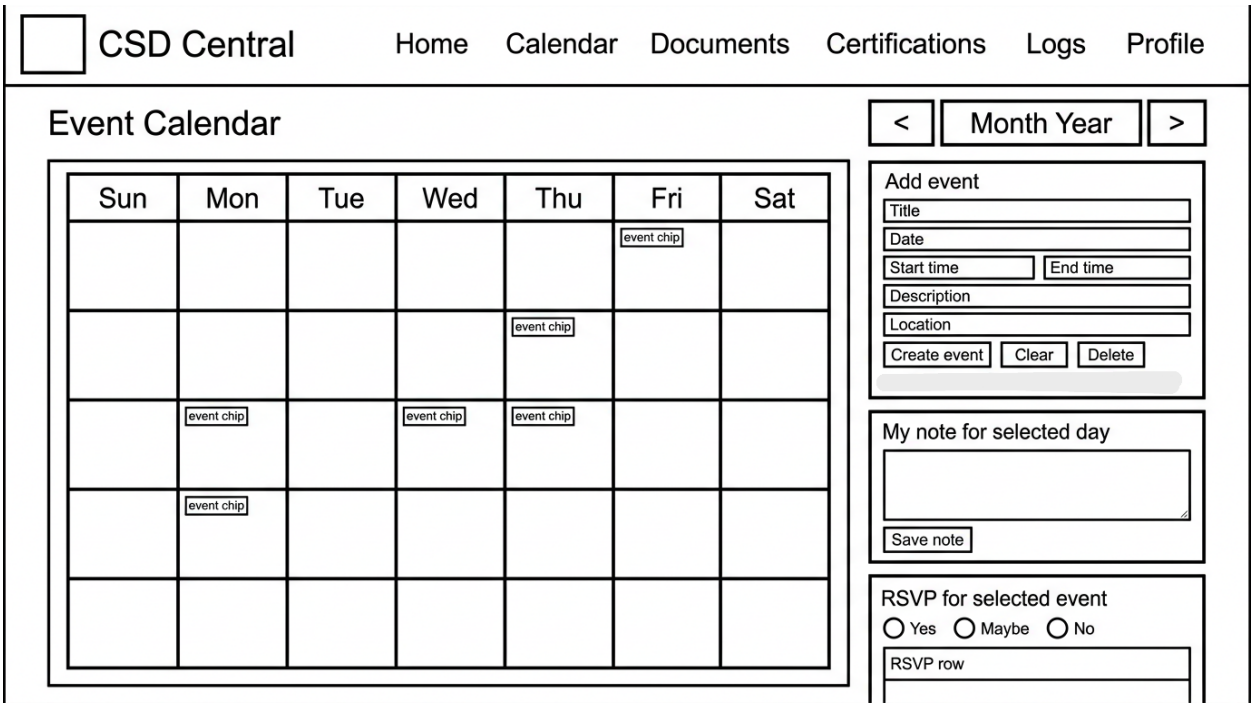


Figure 5: UI Mockup: Event Calendar

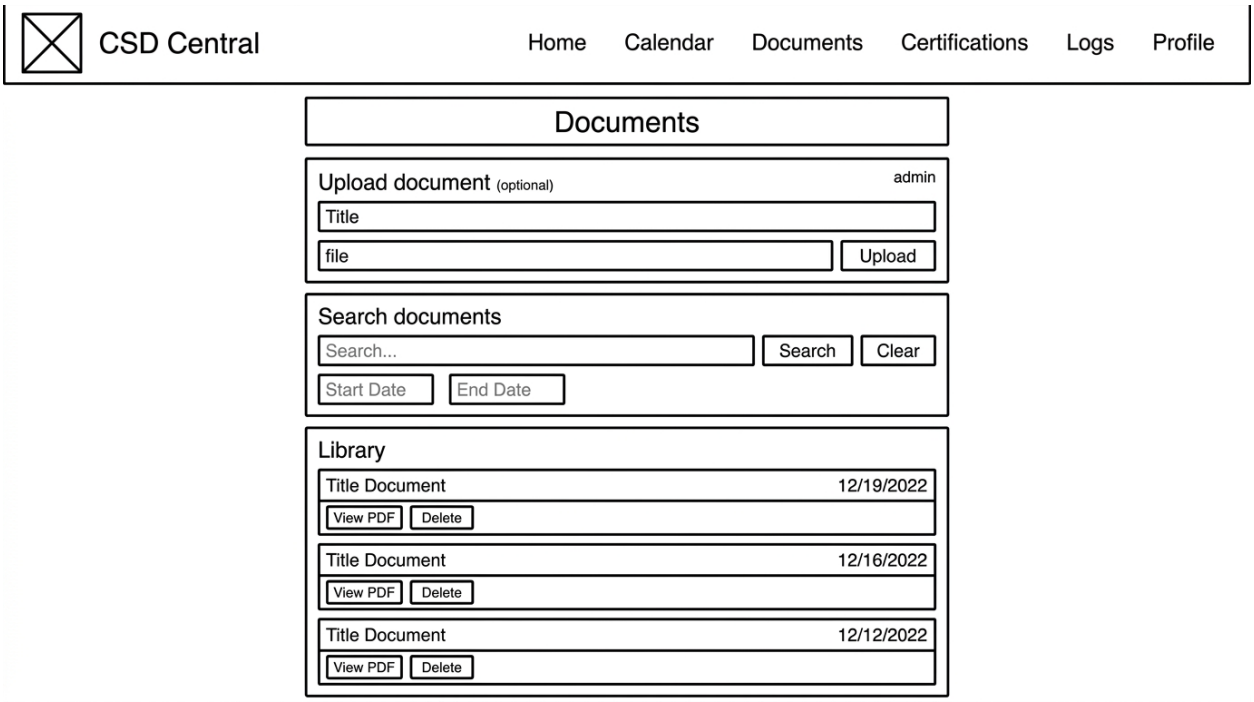


Figure 6: UI Mockup: Certification Upload

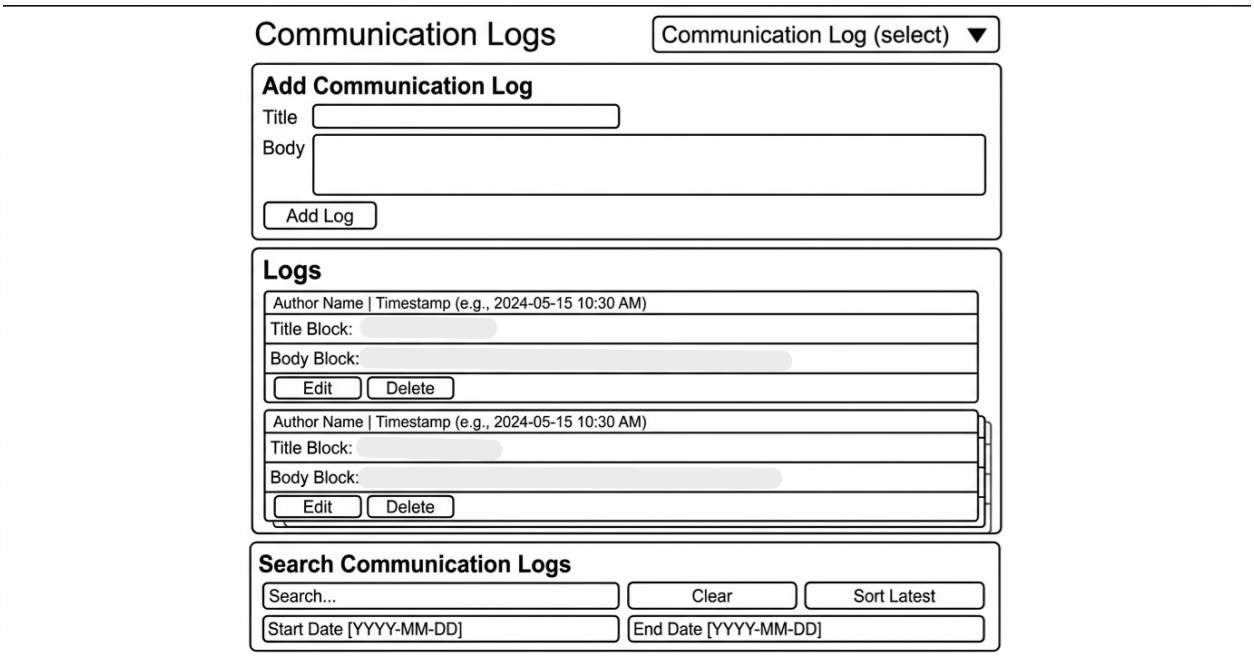


Figure 7: UI Mockup: Training Logs

5 Iterations

5.1 Iteration Planning

Table 4 shows the iteration planning.

It.	Dates	Stories	Points Planned	Points Done
1	01/27 - 02/10	S19 Creating Users, S20 Creating Dog Account, S21 Admin Control	12	12
2	02/10 - 02/24	S9 Upload Certifications, S10 View Certifications, S11 Manage Certifications, S12 Admin Viewing Certifications, S7 Live Calendar	12	0
3	02/24 - 03/17	S8 Event Notes, S5 Adding Events to Calendar, S12 Certifications Filter/Search, S14 Communication Logs	18	0
4	03/17 - 03/31	S4 Event RSVP, S2 Viewing Event RSVPs, S15 Training Logs, S16 Adding Dog to Training Log, S18 Logs Filter/Search, S17 Editing Logs	19	0
5	03/31 - 04/14	S1 Viewing Upcoming Events, S6 Homepage View Event, S3 View Weather	06	0
		Total:	67	12

Table 4: Iteration Planning for Incremental Deliveries

5.2 Iteration/Sprint 1

5.2.1 Planning

We planned on working on S19 creating users, S20 creating dog accounts, and S21 admin control. We wanted to complete user accounts to then build features on top of them. Creating the authentication framework for the project was necessary in order to continue working on future user stories since the project heavily relies on data integrity and security.

5.2.2 Work Done

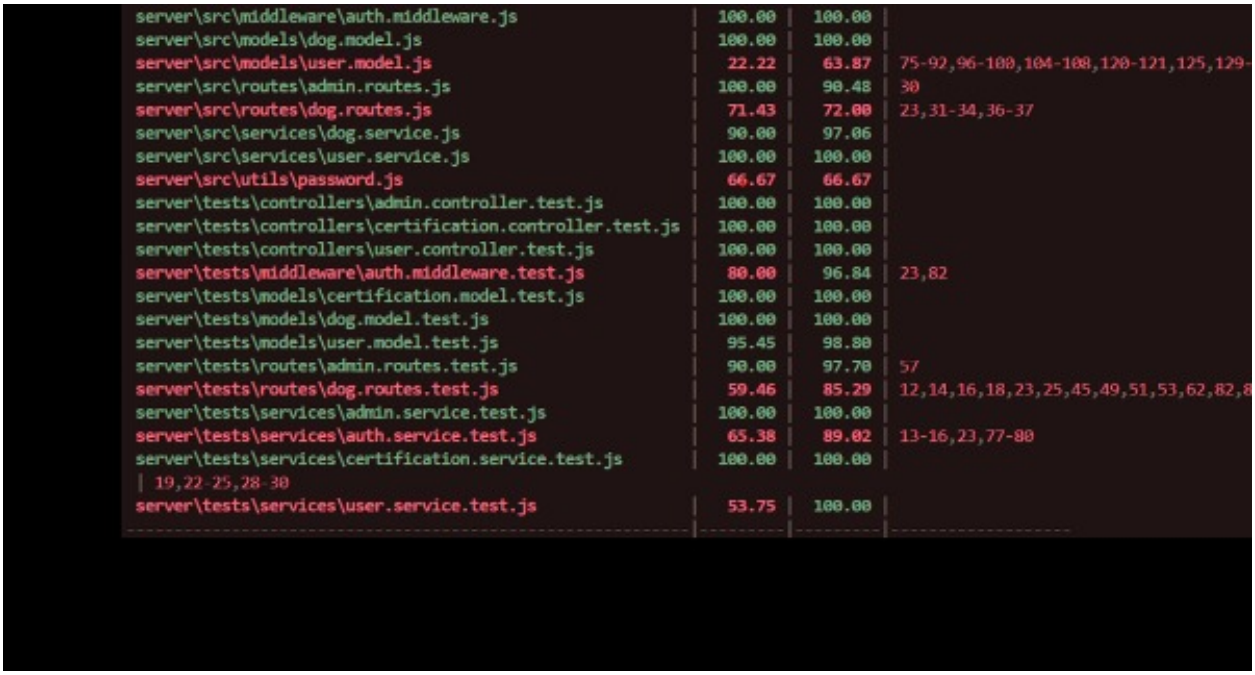
We completed all planned tasks and the authentication framework. Users can now create accounts and dog accounts while admins can have escalated privileges to oversee accounts. Authentication was made using JWT tokens and protected routes.

Ekto: S20 Dog Accounts, S21 Admin Control

Alberto: S19 User Accounts, S21 Admin Control

5.2.3 Testing Coverage

Figure 2 shows the test coverage



server\src\middleware\auth.middleware.js	100.00	100.00	
server\src\models\dog.model.js	100.00	100.00	
server\src\models\user.model.js	22.22	63.87	75-92,96-100,104-108,120-121,125,129-
server\src\routes\admin.routes.js	100.00	90.48	30
server\src\routes\dog.routes.js	71.43	72.00	23,31-34,36-37
server\src\services\dog.service.js	90.00	97.06	
server\src\services\user.service.js	100.00	100.00	
server\src\utils\password.js	66.67	66.67	
server\tests\controllers\admin.controller.test.js	100.00	100.00	
server\tests\controllers\certification.controller.test.js	100.00	100.00	
server\tests\controllers\user.controller.test.js	100.00	100.00	
server\tests\middleware\auth.middleware.test.js	80.00	96.84	23,82
server\tests\models\certification.model.test.js	100.00	100.00	
server\tests\models\dog.model.test.js	100.00	100.00	
server\tests\models\user.model.test.js	95.45	98.80	
server\tests\routes\admin.routes.test.js	90.00	97.70	57
server\tests\routes\dog.routes.test.js	59.46	85.29	12,14,16,18,23,25,45,49,51,53,62,82,8
server\tests\services\admin.service.test.js	100.00	100.00	
server\tests\services\auth.service.test.js	65.38	89.02	13-16,23,77-80
server\tests\services\certification.service.test.js	100.00	100.00	
19,22-25,28-30			
server\tests\services\user.service.test.js	53.75	100.00	

Figure 8: Iteration 1 test coverage report

We achieved high coverage in this iteration but missed a few areas due to some refactoring that had to be done last minute. Tests will be refactored to reflect these changes to achieve a high coverage percentage.

5.2.4 Retrospective & Reflection

Some issues we had with this iteration was when it came to creating the authentication framework with JWT and MongoDB. It just took a bit longer to figure out and delayed some progress but was necessary to have the features fully functional.

5.3 Iteration/Sprint 2

5.3.1 Planning

Stories planned for this iteration are S9 Upload Certifications, S10 View Certifications, S11 Manage Certifications, S12 Admin Viweing Certifications, S7 Live Calendar. We decided to focus on one big feature and we decided on the certification feature since it dealt with file uploads. We wanted to make sure to get as much done regarding this feature

5.3.2 Work Done

No story was fully completed but a lot of work was done towards completion. The certification feature is almost completed alongside its corresponding user stories. It seems like not much work was done but it is mostly complete with some bugs. The live calendar needs a lot of work on the front-end but backend functionality should be good.

We also implemented GitHub actions for better version control and tracking in the future. This is used to track features and code integrity better while encouraging the use of pull requests to organize commits.

Also this iteration may show little to no commits, because branches were not merged to main.

Ekto: S7 Live Calendar

Alberto: S9 Upload Certifications, S10 View Certifications, S11 Manage Certi- fications, S12 Admin Viweing Certifications

It.	Dates	Stories	Points Planned	Points Done
1	01/27 - 02/10	S19 Creating Users, S20 Creating Dog Account, S21 Admin Control	12	12
2	02/10 - 02/24	S9 Upload Certifications, S10 View Certifications, S11 Manage Certifications, S12 Admin Viewing Certifications, S7 Live Calendar	12	0
3	02/24 - 03/17	S8 Event Notes, S5 Adding Events to Calendar, S12 Certifications Filter/Search, S14 Communication Logs	18	0
4	03/17 - 03/31	S4 Event RSVP, S2 Viewing Event RSVPs, S15 Training Logs, S16 Adding Dog to Training Log, S18 Logs Filter/Search, S17 Editing Logs	19	0
5	03/31 - 04/14	S1 Viewing Upcoming Events, S6 Homepage View Event, S3 View Weather	06	0
		Total:	67	12

Table 5: Iteration Planning for Incremental Deliveries

5.3.3 Testing Coverage

No testing was done during this iteration due to the incomplete nature of the work done. Will be completed in the future iterations.

5.3.4 Retrospective & Reflection

A huge pitfall we had during this iteration was the fact that it was during our midterms causing a halt in progress. It severely slowed progress towards our goal for this iteration.

5.4 Iteration/Sprint 3

5.4.1 Planning

We planned on working on S8 Event Notes, S5 Adding Events to Calendar, S12 Certifications Filter/Search, and S14 Communicaton Logs this iteration which ended up to be 18 story points. This builds on the certifications and calendar feature from Iteration 2. Working on the filter/search this far into the project is also helpful considering the potential to be complex. Further iterations focus on building on previous features and anything left incomplete.

5.4.2 Work Done

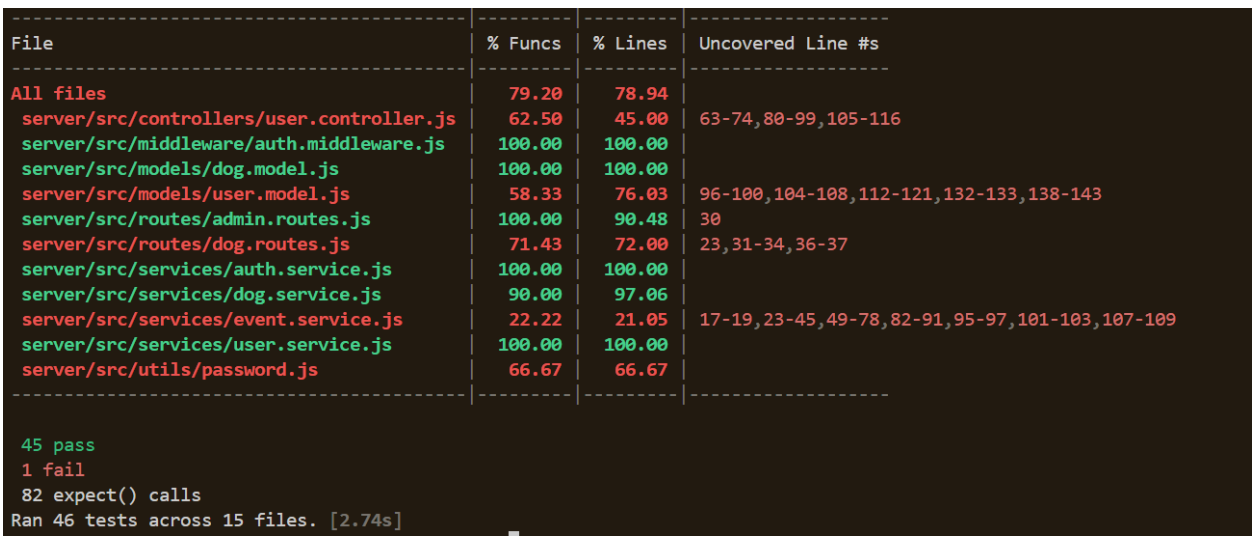
S8 Event Notes, S5 Adding Events to Calendar, and S14 Communicaton Logs were completed this iteration on top of previous stories such as S7 Live Calendar from Iteration 2. S14 Communication Logs was a completely new feature and was completed, testing and final frontend UI is incomplete. We are waiting for client feedback before proceeding. Any feedback will be applied to our work accordingly.

Ekto: S7 Live Calendar, S8 Event Notes, S5 Adding Events to Calendar

Alberto: S9 Upload Certifications, S10 View Certifications, S11 Manage Certifications, S12 Admin Viewing Certifications, S12 Certifications Filter/Search, S14 Communication Logs

It.	Dates	Stories	Points Planned	Points Done
1	01/27 - 02/10	S19 Creating Users, S20 Creating Dog Account, S21 Admin Control	12	12
2	02/10 - 02/24	S9 Upload Certifications, S10 View Certifications, S11 Manage Certifications, S12 Admin Viewing Certifications, S7 Live Calendar	12	9
3	02/24 - 03/17	S8 Event Notes, S5 Adding Events to Calendar, S12 Certifications Filter/Search, S14 Communication Logs	18	13
4	03/17 - 03/31	S4 Event RSVP, S2 Viewing Event RSVPs, S15 Training Logs, S16 Adding Dog to Training Log, S18 Logs Filter/Search, S17 Editing Logs	19	0
5	03/31 - 04/14	S1 Viewing Upcoming Events, S6 Homepage View Event, S3 View Weather	06	0
		Total:	67	34

Table 6: Iteration Planning for Incremental Deliveries



File	% Funcs	% Lines	Uncovered Line #s
All files	79.20	78.94	
server/src/controllers/user.controller.js	62.50	45.00	63-74,80-99,105-116
server/src/middleware/auth.middleware.js	100.00	100.00	
server/src/models/dog.model.js	100.00	100.00	
server/src/models/user.model.js	58.33	76.03	96-100,104-108,112-121,132-133,138-143
server/src/routes/admin.routes.js	100.00	90.48	30
server/src/routes/dog.routes.js	71.43	72.00	23,31-34,36-37
server/src/services/auth.service.js	100.00	100.00	
server/src/services/dog.service.js	90.00	97.06	
server/src/services/event.service.js	22.22	21.05	17-19,23-45,49-78,82-91,95-97,101-103,107-109
server/src/services/user.service.js	100.00	100.00	
server/src/utils/password.js	66.67	66.67	

```
 45 pass
 1 fail
 82 expect() calls
Ran 46 tests across 15 files. [2.74s]
```

Figure 9: Iteration 3 test coverage report

5.4.3 Testing Coverage

Figure 9 shows the test coverage

This is our current coverage. It is considerably low due to the decision to focus on completing features to receive client feedback. Once this feedback is received, more tests will be written to reflect any necessary changes made based on the feedback.

5.4.4 Retrospective & Reflection

Pitfalls this iteration were that we had to move Iteration 2 stories into this iteration and were unable to fix the certification upload feature. This will be moved to Iteration 4 alongside certification search features. In order to avoid issues like this, we need to take into consideration the potential amount of work it could take to complete certain user stories. This has been an issue plaguing our development cycles but it greatly affected this feature due to the nature of the problem.

5.5 Iteration/Sprint 4

5.5.1 Planning

S18 Logs Filter/Search and S12 Certifications Filter/Search are both being pushed to Iteration 5, since they are similar and should be completed once prerequisite features are polished. For this iteration we planned to complete S4 Event RSVP, S2 Viewing Event RSVPs, S15 Training Logs, S16 Adding Dog to Training Log, S17 Editing Logs.

5.5.2 Work Done

All planned features were completed. Additionally, we were finally able to debug S9 Upload Certifications, and complete that user story.

Ekto: S4 Event RSVP, S2 Viewing Event RSVPs, S15 Training Logs, S16 Adding Dog to Training Log

Alberto: S17 Editing Logs, S9 Upload Certifications

			Points	
It.	**Dates**	**Stories**	**Planned**	**Done**
1	01/27 - 02/10	S19 Creating Users, S20 Creating Dog Account, S21 Admin Control	12	12
2	02/10 - 02/24	S9 Upload Certifications, S10 View Certifications, S11 Manage Certifications, S12 Admin Viewing Certifications, S7 Live Calendar	12	12
3	02/24 - 03/17	S8 Event Notes, S5 Adding Events to Calendar, S14 Communication Logs	13	13
4	03/17 - 03/31	S4 Event RSVP, S2 Viewing Event RSVPs, S15 Training Logs, S16 Adding Dog to Training Log, S17 Editing Logs	14	14
5	03/31 - 04/14	S1 Viewing Upcoming Events, S6 Homepage View Event, S3 View Weather, S18 Logs Filter/Search, S12 Certifications Filter/Search	16	0
		Total:	67	51

Table 7: Iteration Planning for Incremental Deliveries

5.5.3 Testing Coverage

Tests were decent this iteration, however a certain amount of them are out of date or refactoring code has caused them to be low or failing.

File	% Funcs	% Lines	Uncovered Line #s
All files	79.74	89.84	
server\src\controllers\search.controller.js	100.00	100.00	
server\src\controllers\user.controller.js	100.00	100.00	
server\src\middleware\auth.middleware.js	50.00	27.50	4-32
server\src\models\dog.model.js	100.00	100.00	
server\src\models\user.model.js	61.54	78.69	96-100,104-108,112-121,125,129-133
server\src\routes\admin.routes.js	100.00	92.50	45,53
server\src\routes\commLog.routes.js	50.00	68.18	15,20,25,27,29,39
server\src\routes\dog.routes.js	100.00	100.00	
server\src\routes\search.routes.js	100.00	100.00	
server\src\routes\trainingLog.routes.js	50.00	71.43	15,19,23,25,32
server\src\services\commLog.service.js	50.00	46.91	9-24,28-40,44-56
server\src\services\dog.service.js	91.67	94.23	25,50
server\src\services\event.service.js	100.00	100.00	
server\src\services\trainingLog.service.js	50.00	47.56	9-35,39-40,44-52,80,83,89,92
server\src\utils\filterQuery.js	66.67	81.58	3,25-30
server\src\utils\searchQuery.js	100.00	100.00	
server\src\utils\sortQuery.js	100.00	100.00	
server\tests\controllers\admin.controller.test.js	100.00	100.00	
server\tests\controllers\certification.controller.test.js	100.00	100.00	
server\tests\controllers\search.controller.test.js	100.00	100.00	
server\tests\controllers\user.controller.extra.test.js	44.14	100.00	
server\tests\controllers\user.controller.test.js	100.00	100.00	
server\tests\middleware\auth.middleware.test.js	75.00	97.12	31,92
server\tests\models\certification.model.test.js	100.00	100.00	
server\tests\models\dog.model.test.js	91.67	100.00	
server\tests\models\user.model.test.js	97.78	99.40	
server\tests\routes\admin.routes.test.js	86.96	96.90	64,96,140
server\tests\routes\commLog.routes.test.js	76.92	100.00	
server\tests\routes\dog.routes.test.js	61.90	86.21	19,21,23,25,30,32,52,56,58,60,69,89,91,93,97,124,126,130,132,141,158,160,162
server\tests\routes\search.routes.test.js	100.00	100.00	
server\tests\routes\trainingLog.routes.test.js	76.92	100.00	
server\tests\services\admin.service.test.js	100.00	100.00	
server\tests\services\auth.service.test.js	51.28	88.89	24-27,32,78-81
server\tests\services\certification.service.test.js	100.00	100.00	
server\tests\services\commLog.service.test.js	70.83	100.00	
server\tests\services\dog.service.test.js	90.91	93.98	26,29-32,35-37
server\tests\services\event.rsvp.service.test.js	60.00	100.00	
server\tests\services\event.service.test.js	55.26	98.47	35-36,58-59,78-79
server\tests\services\search.service.test.js	0.00	21.65	69-74,78-299
server\tests\services\trainingLog.service.test.js	70.83	100.00	
server\tests\services\user.service.test.js	28.07	61.68	63-65,69-73,77-94,98-114,120-126,130-132,136-138,142-145,149-152
server\tests\utils\filterQuery.test.js	100.00	100.00	
server\tests\utils\searchQuery.test.js	100.00	100.00	
server\tests\utils\sortQuery.test.js	100.00	100.00	

Figure 10: Iteration 4 test coverage report

5.5.4 Retrospective & Reflection

A real challenge is getting filters and search query working for both certifications and logs, so we have decided to push it to the last short iteration. The uploading certifications issue was finally solved, we underestimated how difficult saving file types to the database could be.

5.6 Iteration/Sprint 5

5.6.1 Planning

We have an additional last-minute feature this iteration, however it is very similar to certifications, so we should be able to complete it. For this iteration we have planned S1 Viewing Upcoming Events, S6 Homepage View Event, S3 View Weather, S18 Logs Filter/Search, S12 Certifications Filter/Search, S22 Adding Documents, S23 Deleting Documents, S24 Viewing Documents, S25 Documents Filter/Search.

5.6.2 Work Done

We were able to finish all user stories and polish the website. The new feature was not too difficult to implement since it was similar to certifications.

Ekto: S1 Viewing Upcoming Events, S6 Homepage View Event, S3 View Weather, S22 Adding Documents, S23 Deleting Documents, S24 Viewing Documents.

Alberto: S18 Logs Filter/Search, S12 Certifications Filter/Search, S25 Documents Filter/Search

It.	Dates	Stories	Points Planned	Points Done
1	01/27 - 02/10	S19 Creating Users, S20 Creating Dog Account, S21 Admin Control	12	12
2	02/10 - 02/24	S9 Upload Certifications, S10 View Certifications, S11 Manage Certifications, S12 Admin Viewing Certifications, S7 Live Calendar	12	12
3	02/24 - 03/17	S8 Event Notes, S5 Adding Events to Calendar, S14 Communication Logs	13	13
4	03/17 - 03/31	S4 Event RSVP, S2 Viewing Event RSVPs, S15 Training Logs, S16 Adding Dog to Training Log, S17 Editing Logs	14	14
5	03/31 - 04/14	S1 Viewing Upcoming Events, S6 Homepage View Event, S3 View Weather, S18 Logs Filter/Search, S12 Certifications Filter/Search, S22 Adding Documents, S23 Deleting Documents, S24 Viewing Documents, S25 Documents Filter/Search	26	26
		Total:	77	77

Table 8: Iteration Planning for Incremental Deliveries

5.6.3 Testing Coverage

File	% Funcs	% Lines	Uncovered Line #s
All files	84.87	86.83	
server/src/controllers/search.controller.js	100.00	100.00	
server/src/controllers/user.controller.js	100.00	100.00	
server/src/middleware/auth.middleware.js	50.00	27.50	4-32
server/src/models/dog.model.js	100.00	100.00	
server/src/models/user.model.js	61.54	78.69	96-100,104-108,112-121,125,129-133
server/src/routes/admin.routes.js	100.00	92.50	45,53
server/src/routes/commLog.routes.js	50.00	68.18	15,20,25,27,29,39
server/src/routes/dog.routes.js	100.00	100.00	
server/src/routes/search.routes.js	100.00	100.00	
server/src/routes/trainingLog.routes.js	50.00	71.43	15,19,23,25,32
server/src/services/auth.service.js	100.00	100.00	
server/src/services/dog.service.js	91.67	94.23	25,50
server/src/services/event.service.js	22.22	20.00	17-19,23-46,50-83,87-96,100-102,106-108,112-114
server/src/services/search.service.js	87.10	97.33	20-21,99,160,192,196
server/src/services/user.service.js	100.00	100.00	
server/src/utils/filterQuery.js	100.00	100.00	
server/src/utils/password.js	100.00	100.00	
server/src/utils/searchQuery.js	100.00	100.00	
server/src/utils/sortQuery.js	100.00	100.00	

Figure 11: Iteration 3 test coverage report

Figure 11 shows the test coverage

This is our test coverage for iteration 5. Managed to achieve 86% line coverage Had issues trying to break past 90% coverage.

5.6.4 Retroespective & Reflection

Adding a last-minute feature was not ideal, but we were able to finish it. We were lucky that it was similar to certifications, and that the other features were fairly simple for this iteration. The weather API was a new experience and took slightly longer to implement than intended. And we were able to finally able complete search functions and filters for logs, certifications, and documents.

6 Final Remarks

6.1 Overall Progress

We were able to able to complete everything, as of this draft submission we are waiting for final feedback of our product, but based off previous feedback they should be pleased.

6.2 Project Reflection

Ekto: This project was a good experience and I think we were able to deliver a good product. Making a product for an actual client was a new experience and it provided lots of valuable lessons in communicating and creating for a client. One problem we faced as partners in the early iterations was communicating, I even ended up overwriting Alberto's work and having to rollback to a previous commit, but after that harsh setback we properly provided updates on work being done and how it was being implemented. I think for a future project I would definitely try to properly estimate the difficulty of certain features being implemented, there were a good amount of user stories that were underestimated in terms of difficulty for me personally. We did fullstack Javascript with Bun, working with Bun was new for me but it did not pose any real issues. However after this project, Javascript has become less favorable for me, it makes me reconsider using Java Spring framework or learning a new framework for future projects. I was aware of the drawbacks of MongoDB for the project, I think it sufficed and got the job done but maybe there are better database options in the future. The Bun testing tool was great to use, was simple to use and displayed results in a very neat manner. When I use Javascript in the future, I think Bun framework would be my first choice. As far as maintainability, we chose a layered backend layout with reasonable file names and clear file structures. Also, any environment can quickly run the project as long as they have a .env file. As I ran into issues debugging Javascript written code, I occasionally used AI to help debug snippets of code that I just could not figure out, it did not always help but offered perspective on what the issue could possibly be.

Alberto: Working with a client to deliver a fully functional application was an interesting experience. I enjoyed it nonetheless, regardless of difficult aspects of the project. This was my first time writing a full-stack Javascript web app using Bun. When I was creating a design and figuring out what technologies to use, I found Bun. We decided to go with a full stack Javascript just to make it more consistent between the fronted and backend. I found Bun interesting because of its built in test suite and its ability to run Typescript and Javascript, mostly interested in its test suite. The experience I gained with system design in this project helped a lot in outside projects where I was able to gather requirements and find the right technologies for the problem. I did end up finding software engineering more enjoyable as a result. Working with a real client also made it interesting since it brought a level of uncertainty and pressure to deliver. This was great for real world experience. Although our project went somewhat smoothly for the most part, I do have some qualms. I wish we could have chosen a different language. I did not find it enjoyable writing in Javascript after a long time working on the project. I also do not enjoy MongoDB. Its great for files and data that needs to be unstructured but its a headache to work with at least for me. You can achieve a lot of what MongoDB has to offer with PostgreSQL and its JSONB data type. Its also open source and I am someone who will advocate for open source when designing software. Though, at the time of designing and starting the project, I was unaware of the JSONB data type and MongoDB works well for the project scope. We did end up completing our project and all of our user stories with no left over work.